

Q1. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

Nine distinct integers are divided into three groups of three integers each. Now, one integer is taken from each of the three groups and the three integers so obtained are added. Each such possible sum is equal to a distinct integer from among 1 to 27. What is the minimum possible value of the greatest of these nine integers?

Q2. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

If the sum of the interior angles of a convex polygon is seven times the sum of its exterior angles, how many sides does the polygon have?

Q3. DIRECTIONS *for question 3:* Select the correct alternative from the given choices.

If S_m represents the sum of the first n terms of the m^{th} arithmetic progression, whose first term is m^2 and common difference is m , then the value of $S_1 + S_2 + S_3 + \dots + S_m$ is

- ☐ a) $\frac{mn(m + 1)}{2}$
- ☐ b) $\frac{mn(2m + 1)}{3}$
- ☐ c) $\frac{mn[3(m + n) + 1]}{6}$
- ☐ d) $\frac{mn(m + 1)(4m + 3n - 1)}{12}$

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

If for all real numbers, a and b , $a * b = 2 - ab$, then how many of the following statements are true?

- I. $(3 * 2) * (2 * 3) = -14$
- II. $(4 * 2) * (2 * 3) = (1 * 6) * (8 * 1)$
- III. $(-1 * 2) * (-2 * 1) = (2 * 8)$
- IV. $(6 * 1) * (3 * 2) = (7 * 3) * (1 * 2)$

Q5. DIRECTIONS *for question 5:* Select the correct alternative from the given choices.

P, Q and R can complete a work in 12, 15 and 20 days respectively. If on any given day only two of them work and the same two people do not work on any two consecutive days, find the minimum number of days in which the work can be done.

☐ a) $8\frac{8}{9}$

☐ b) $8\frac{4}{9}$

☐ c) 6

☐ d) 7

DIRECTIONS for questions 6 and 7: Answer the questions on the basis of the information given below.

This morning, Madhu, the local milkman, was in a hurry to leave home for selling milk. In his hurry he asked Bhola, his servant, to pour half of the water from the water drum into the milk drum containing pure milk and then to bring him a full can of the diluted milk for selling. However, Bhola got confused and he poured half the milk from the milk drum into the water drum and brought out a full can of the contents from the latter for his master to sell. The concentration of the milk in the can that Bhola brought was $1/n$ times that intended by Madhu.

Q6. DIRECTIONS for questions 6 and 7: Select the correct the alternative from the given choices.
If $n = 3$, what would be the concentration of the milk had Madhu mixed the entire contents of the milk drum and the water drum?

- ☐ a) $33\frac{1}{3}\%$
- ☐ b) **25%**
- ☐ c) 20%
- ☐ d) Cannot be determined

DIRECTIONS *for questions 6 and 7:* Answer the questions on the basis of the information given below.

This morning, Madhu, the local milkman, was in a hurry to leave home for selling milk. In his hurry he asked Bhola, his servant, to pour half of the water from the water drum into the milk drum containing pure milk and then to bring him a full can of the diluted milk for selling. However, Bhola got confused and he poured half the milk from the milk drum into the water drum and brought out a full can of the contents from the latter for his master to sell. The concentration of the milk in the can that Bhola brought was $1/n$ times that intended by Madhu.

Q7. DIRECTIONS *for questions 6 and 7:* Select the correct the alternative from the given choices.

Find n , given that initially the total quantity of pure milk in the milk drum was 4.5 times that of water in the water drum.

- ☐ a) 1.3
- ☐ b) 1.7
- ☐ c) 2
- ☐ d) Cannot be determined

Q8. DIRECTIONS *for question 8:* Type in your answer in the input box provided below the question.

A natural number N has seven factors greater than $\sqrt{N}$. If $\sqrt{N}$ is also a natural number, at most how many factors can $N^{3/2}$ have?

Q9. DIRECTIONS *for questions 9 to 11:* Select the correct alternative from the given choices.

If two percent of the population of constituency A are octogenarians, whereas six percent of the population of constituency B are octogenarians, and the population of B is three times that of A, what percent of the population of A and B together are octogenarians?

- ☐ a) 4%
- ☒ b) 5% ✓ Your answer is correct
- ☐ c) $3\frac{1}{4}\%$
- ☐ d) $4\frac{1}{3}\%$

Q10. DIRECTIONS *for questions 9 to 11:* Select the correct alternative from the given choices.

On a circular track, with center O, A starts at a point P and runs in the clockwise direction. At the same time, B starts at Q and runs in the anticlockwise direction. The angle $\angle POQ$, measured in the clockwise direction, is 270° . The distance covered by B by the time he meets A for the first time is equal to the distance covered by A between the first and second meetings. What fraction of the track length does A cover by the first meeting?

☐ a) $\frac{1}{3}$

☐ b) $\frac{9}{28}$

☐ c) $\frac{2}{9}$

☐ d) $\frac{9}{26}$

Q11. DIRECTIONS for questions 9 to 11: Select the correct alternative from the given choices.

If $f(x) = \frac{x}{1+x^2}$ and $g(x) = \frac{e^{-x}}{1+[x]}$, where $[x]$ is the greatest integer less than or equal to x , which of the following is/are true?

- I. domain of $(f+g) = \mathbb{R} - (-2, -1]$
- II. domain of $(f+g) = \mathbb{R} - [-1, 0)$
- III. $[\text{range of } f] \cap [\text{range of } g] = \left[-2, \frac{1}{2}\right]$
- IV. $[\text{range of } f] \cap [\text{range } g] = \left[\frac{-1}{2}, \frac{1}{2}\right] - \{0\}$

- ☐ a) **Both II and IV**
- ☐ b) **Both I and III**
- ☐ c) Both I and IV
- ☐ d) Both II and III

Q12. DIRECTIONS *for questions 12:* The question is followed by two statements, I and II. You have to decide whether the information provided in the statements is sufficient for answering the question and then select the correct alternative from the given choices.

Is $|y - 2| < 1$?

I. $|y| > 1$

II. $|y - 1| < 2$

- ☐ a) The question can be answered by using only one of the statements.
- ☐ b) The question can be answered by using either statement alone.
- ☐ c) The question can be answered by using both the statements together but cannot be answered by using either statement alone.
- ☐ d) The question cannot be answered even when both the statements are used together.

Q13. DIRECTIONS *for question 13:* Select the correct alternative from the given choices.

A cylindrical block of wood, of radius 14 cm and height 20 cm, is cut into four identical pieces by making two cuts perpendicular to its base. If three of the four pieces are again glued back together in their original configuration, then what is the total surface area of this new block of wood? (Consider $\pi = \frac{22}{7}$)

- ☐ a) 3728 sq.cm
- ☐ b) 2704 sq.cm
- ☐ c) 3608 sq.cm
- ☒ d) 2804 sq.cm ✓ Your answer is correct

Q14. DIRECTIONS *for question 14:* Type in your answer in the input box provided below the question.

If the quadratic equations $x^2 - ax + 3 = 0$ and $x^2 + ax - 5 = 0$ have one positive root in common, find the value of a .

Q15. DIRECTIONS *for questions 15 to 17:* Select the correct alternative from the given choices.

On January 1st, 2003, the average age of 11 members of a cricket team was 30 years. After four years, a player X from the team was replaced by another player Y. After one more year, player Y was replaced by another player Z, as a result of which the average age of the team then became $34\frac{4}{11}$ years. If the age of player Y is 4 years less than that of player X, then the age of player Z is how many years more/less than that of player Y?

- ☐ a) **3 years less**
- ☐ b) **2 years more**
- ☐ c) 4 years less
- ☐ d) **Cannot be determined**

Q16. DIRECTIONS *for questions 15 to 17:* Select the correct alternative from the given choices.

The number of quadrilaterals that can be formed by joining the corners of a regular nonagon, such that none of the quadrilaterals have any side in common with the sides of the nonagon, is

☐ a) 7.

☐ b) 8.

☐ c) 9.

☐ d) 6.

Q17. DIRECTIONS *for questions 15 to 17:* Select the correct alternative from the given choices.

Rohan went to a casino to play a card game. At the end of each round of the game, he doubled the amount with him and gave ₹2,048 to his friend. After playing ten rounds, he was left with no money. What was the amount he started with?

- ☐ a) ₹3,071
- ☐ b) ₹4,098
- ☐ c) ₹2,046
- ☐ d) ₹2,047

Q18. DIRECTIONS *for question 18:* Type in your answer in the input box provided below the question.

Raghu, Ram and Rajan work in ABC Pvt. Ltd. If their monthly salaries are in the ratio 2 : 3 : 5, and sum of their monthly salaries is ₹720000, find Raghu's monthly salary (in ₹).

Q19. DIRECTIONS *for question 19:* Select the correct alternative from the given choices.

Nine parallel chords are drawn in a circle of diameter 10 cm. If the distance between any two of the adjacent chords is 1 cm, which of the following statements is always true?

- ☐ a) **One of the chords is the diameter of the circle.**
- ☐ b) **At least two of the chords must be of equal length.**
- ☐ c) The difference between the lengths of any two adjacent chords on the same side of the diameter is greater than 1 cm.
- ☐ d) **None of the above**

Q20. DIRECTIONS *for questions 20:* Type in your answer in the input box provided below the question.

If $a_1, a_2, a_3, \dots, a_{10}$ are in A.P, $h_1, h_2, h_3, \dots, h_{10}$ are in H.P., and $a_1 = h_1 = 4$, while $a_{10} = h_{10} = 5$, then $a_4 h_7 =$

Q21. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

If the difference between a two-digit number and the sum of its digits is X times the sum of its digits, then the difference between the two-digit number formed by interchanging the digits of the number and the sum of its digits is

- ☐ a) $(9 - X)$ times the sum of its digits.
- ☐ b) $(9 + X)$ times the sum of its digits.
- ☐ c) $(9 + X)$ times the difference of its digits.
- ☐ d) $(9 - X)$ times the difference of its digits.

Q22. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

Given $|p| \geq 1$, what is the area of the region bounded by the lines $|x| = |p|$, $x - y = 0$ and the x -axis?

- ☐ a) $2p^2$
- ☐ b) p^2
- ☐ c) $p^2/2$
- ☐ d) $p^2 + 1$